

Query Scheduling Optimization in IVF-based Vector Search

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Abstract

Vector search is a core component of modern AI systems, yet most optimization efforts focus on index structure rather than query execution. This paper investigates whether throughput can be improved by batching concurrent queries over an IVF index to exploit overlap in the inverted lists they probe, without modifying the index itself. We compare two batching strategies, Batch(MV) and Batch(MM), against a sequential baseline under random and clustered query workloads. Experiments on SIFT1M confirm that batching consistently improves throughput over sequential execution: Batch(MV) outperforms Sequential by up to 19% on random queries, while Batch(MM) achieves up to 40% gains on clustered queries, with all schedulers preserving identical recall. The preferred scan mode shifts with the query distribution over IVF clusters, and batch size further modulates the relative performance between MV and MM. A microbenchmark isolates the mechanism underlying these differences and supports practical guidance on applying batch scheduling in IVF-based vector search.

Keywords

vector search, IVF, batch scheduling, GEMM, query execution

1 Introduction

Vector search is central to a wide range of AI applications, including retrieval-augmented generation, semantic search, and recommendation systems. Given a query vector, the system must efficiently return the k nearest neighbors from a large corpus of stored vectors. At scale, such systems must sustain high query throughput while keeping tail latency bounded.

The dominant approach to improving vector search performance is to optimize the index structure. Hierarchical Navigable Small World graphs (HNSW) [4], product quantization [2], and IVF variants [3] have each offered significant gains in recall-efficiency trade-offs. By contrast, the query execution layer—how individual queries are dispatched and processed at runtime—has received comparatively less attention.

This work takes the index as fixed and asks: **can throughput be improved by reorganizing how concurrent queries are executed?**

We focus on IVF (Inverted File Index), which partitions the vector space into clusters, each maintaining an inverted list of assigned vectors. To answer a query, the engine identifies the nearest centroids and scans the corresponding lists. Crucially, when multiple queries are processed together, many of them probe overlapping lists. If the engine exploits this overlap—loading each list once and computing distances for all queries that need it—the total number of memory loads decreases substantially.

Some recent systems have also focused on this batching strategy. Milvus groups concurrent queries into cache-conscious blocks [7];

HQI batches queries over shared clusters for high-throughput vector search [6]; Quake measures batching effects in the context of adaptive indexing [5]; and GPU-native IVF work uses batching to expose GEMM-friendly computation and coalesced memory access [1]. These systems demonstrate the value of batching, but they do not isolate when GEMM-based list scanning should replace per-query GEMV under a fixed index, nor how the query distribution over IVF clusters and batch size govern that choice.

To study these conditions, we vary three aspects of the execution setting. First, we compare random queries with clustered queries because they induce different degrees of list overlap. Second, we compare two scan modes: Batch(MV), which scans each list query by query, and Batch(MM), which stacks queries sharing a list into a matrix-matrix multiplication. Third, we vary batch size to determine how much cross-query sharing is needed before batching and GEMM become effective.

Based on this setup, we ask four research questions:

- **RQ1.** Does time-window batching improve throughput over sequential execution, under both random and clustered workloads?
- **RQ2.** How does scan mode choice (MV vs MM) affect performance under random versus clustered query distributions?
- **RQ3.** How does batch size affect the performance of batch schedulers?
- **RQ4.** What is the underlying mechanism that governs the relative performance of the two scan modes?

We organize the evaluation into three experiments: a main experiment comparing all schedulers under fixed parameters, a batch-size sweep, and a kernel microbenchmark. We find that batching can outperform sequential execution, that the preferred scan mode changes with query distribution and batch size, and that L —the number of queries sharing each inverted list per batch—is the dominant variable behind the MV–MM trade-off. These findings lead to practical guidelines for choosing batch scheduling strategies under a fixed IVF index.

2 Method

2.1 System Design

2.1.1 IVF Engine. Our engine follows the IVF design from Faiss [3]: base vectors are partitioned into 256 clusters via k -means, each cluster maintaining an inverted list. We use Faiss for centroid training but replace `index.search()` with a custom single-threaded implementation. Faiss’s search is a stateless black box that processes each query independently, making cross-query optimizations invisible. Our engine exposes centroid lookup and list scanning as separate interfaces, giving the scheduler direct control over execution order and batch composition.

Table 1: Scheduler configurations.

Scheduler	Scan mode	Batching policy
Sequential	Single-query GEMV per list	None
Batch(MV)	Per-query GEMV per list	Time + batch size
Batch(MM)	GEMM per list	Time + batch size

2.1.2 Per-List Batch Scan. When a batch of queries is dispatched together, multiple queries often probe the same inverted list. The per-list batch scan exploits this by iterating over inverted lists in list-major order: for each list, distances are computed for all queries that probe it before moving to the next. Each list is therefore loaded from memory exactly once per batch, regardless of how many queries share it. This “load once, serve many” principle mirrors cooperative scan techniques in column-store databases [8].

We define L as the number of queries within a batch that probe the same inverted list. Under sequential execution, $L = 1$ always. Under time-window batching, L grows with batch size and workload query distribution over IVF clusters, and directly determines which scan mode is more efficient.

2.1.3 Scan Modes. Two implementations of the per-list distance computation are provided, differing in how they handle the L queries sharing each list:

Batch(MV). Distances are computed for each of the L queries individually via a matrix-vector multiply ($\text{vecs} @ \mathbf{q}$). Arithmetic intensity is constant at 0.5 FLOP/byte, independent of L .

Batch(MM). All L queries are stacked into a query matrix and distances are computed via a single matrix-matrix multiply ($\text{vecs} @ \mathbf{Q} \cdot \mathbf{T}$). Arithmetic intensity scales as $L/2$ FLOP/byte, making MM increasingly efficient as L grows. Prior work on GPU-native IVF search has noted that batching queries enables GEMM-style execution and coalesced memory access [1]; here we characterize the crossover on CPU. GEMM carries non-trivial setup overhead at small L , degrading performance relative to MV. The crossover point is hardware-specific and is measured empirically in Section 3.3.

2.1.4 Schedulers. Three configurations are evaluated (Table 1).

Time-window batching accumulates arriving queries until either a time limit Δt elapses or a maximum batch size MaxBS is reached, then dispatches all accumulated queries as a single batch. The dual trigger bounds both waiting time at low load and memory pressure at high load.

2.2 Experiment Design

2.2.1 Dataset. All experiments use the SIFT1M benchmark [2]: one million 128-dimensional SIFT descriptors, evaluated under L2 distance, with 10,000 held-out test queries and precomputed ground truth. All vectors reside in memory; no disk I/O occurs during search.

2.2.2 Workloads. Query arrivals are simulated at a target rate of 2,000 QPS. Two workloads are evaluated:

Random. All 10,000 SIFT test queries are used in their original order. Each query probes a largely disjoint set of 8 inverted lists, keeping L small. This represents an adversarial case for batching:

list-reuse benefit is low, and any overhead from batch management directly reduces throughput.

Clustered. Queries are filtered to those whose nearest centroid falls among the 10 most-queried centroids, yielding 791 unique queries. These are sorted by primary centroid and then tiled (round-robin repetition) to reach 10,000 arrivals. Sorting by centroid ensures that consecutive arrivals within each time window belong to the same cluster, maximising L per list per batch. All 791 queries are real SIFT test vectors with valid ground truth, so Recall@10 is well-defined.

2.2.3 Device. All experiments are conducted on a MacBook Pro with an Apple M3 Pro chip and 18 GB unified memory. The engine runs single-threaded throughout to isolate scheduling effects from multi-core parallelism.

2.2.4 Experimental Protocol. We use three experiments to answer the research questions introduced in Section 1.

Experiment 1 — Main Experiment (RQ1, RQ2). All three schedulers are evaluated on both workloads under fixed parameters ($n_{\text{clusters}} = 256$, $n_{\text{probe}} = 8$, $k = 10$, $\Delta t = 5$ ms, MaxBS = 128) across 5 independent runs. Summary statistics are reported as mean $\pm$ std; P99 latency is the median across runs.

Experiment 2 — Batch Size Parameter Sweep (RQ3). A grid search is conducted once on both workloads:

$$\begin{aligned} \Delta t \text{ (ms)} &\in \{0.5, 1, 2, 5, 10, 20, 50\}, \\ \text{MaxBS} &\in \{32, 64, 128, 256\}. \end{aligned}$$

Experiment 3 — Microbenchmark (RQ4). The MV and MM kernels are evaluated in isolation on a synthetic inverted list across a range of L values, with 50 repetitions per point.

3 Results

We present results in three stages. First, the main experiment demonstrates the core throughput claims under fixed scheduler parameters. Second, the batch size parameter sweep shows how batch size modulates the relative performance of MV and MM. Third, the microbenchmark isolates L as the underlying mechanism explaining both.

3.1 Main Experiment

All three schedulers are evaluated at fixed parameters ($\Delta t = 5$ ms, MaxBS = 128) across 5 independent runs.

3.1.1 Performance Overview. Figure 1 shows throughput across all three schedulers under both workloads. The two lines cross between Batch(MV) and Batch(MM): on the random workload, throughput peaks at Batch(MV) then falls for MM; on the clustered workload, it rises monotonically with MM taking the lead. This crossing is the central result of the paper.

Three findings stand out. First, batching can outperform Sequential: Batch(MV) achieves +19% on random, and Batch(MM) achieves +40% on clustered. Second, the relative ordering of MV and MM is reversed across workloads—MV leads on random while MM leads on clustered. Third, these throughput gains come with a latency trade-off from queuing: average latency rises from ~ 0.5 ms (Sequential) to 58–105 ms for batch schedulers. Recall@10 remains identical across all schedulers within each workload, confirming

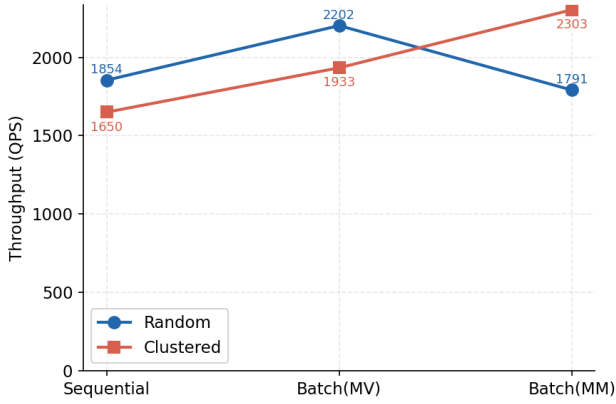


Figure 1: Throughput (QPS) across schedulers for random and clustered workloads.

Table 2: Throughput and Recall@10 (mean $\pm$ std, 5 runs). $\Delta t = 5$ ms, MaxBS = 128.

Scheduler	QPS	Avg (ms)	P95 (ms)	P99 (ms)	Recall @10
<i>Random workload</i>					
Sequential	1,854 $\pm$ 58	0.54	0.74	0.81	0.957
Batch(MV)	2,202 $\pm$ 23	74.34	106.79	114.67	0.957
Batch(MM)	1,791 $\pm$ 87	104.69	134.60	152.89	0.957
<i>Clustered workload</i>					
Sequential	1,650 $\pm$ 33	0.61	0.78	0.88	0.981
Batch(MV)	1,933 $\pm$ 39	96.19	127.30	139.06	0.981
Batch(MM)	2,303 $\pm$ 111	58.09	87.15	106.98	0.981

Table 3: Mean per-list reuse metrics across 5 runs.

Scheduler	Lists/query	L mean	L P50	L P95
<i>Random workload</i>				
Sequential	8.00	1.0	1	1
Batch(MV)	2.23	3.6	3	8
Batch(MM)	1.92	4.2	4	10
<i>Clustered workload</i>				
Sequential	8.00	1.0	1	1
Batch(MV)	0.55	14.6	5	64
Batch(MM)	0.65	12.4	5	52

that the gains arise from execution reorganization rather than approximation.

3.1.2 How Many Queries Share Each Inverted List (L). Sequential always loads 8 lists per query with $L = 1$. On the random workload, batching reduces unique list loads per query to 2.23 (3.6 $\times$ reduction), with mean $L = 3.6$ –4.2. On the clustered workload, this reduction

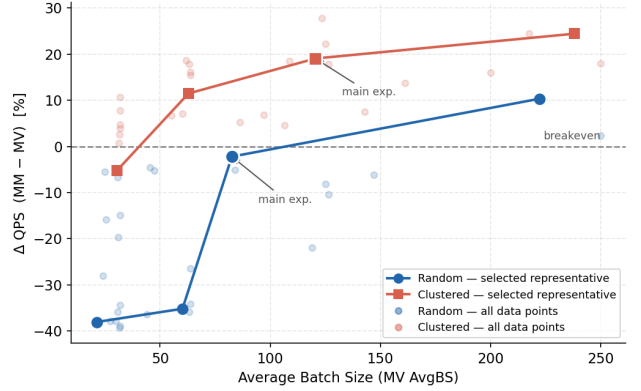


Figure 2: Batch size sweep: relative QPS of MM vs MV across random and clustered workloads.

reaches 0.55 lists per query, with mean $L = 12.4$ –14.6. The contrast in L across the two workloads points to the underlying mechanism distinguishing MV and MM performance—addressed directly by the microbenchmark.

3.2 Batch Size Parameter Sweep

We sweep Δt and MaxBS on both workloads to examine how batch size affects the MV–MM trade-off. Four representative configurations are selected per workload. The sequential baseline achieves 1,937 QPS on random and 1,650 QPS on clustered. In Tables 4 and 5, Δ QPS is computed as $(MM - MV)/MV$.

3.2.1 Random Workload. As Δt and MaxBS grow, L increases and MM’s disadvantage narrows and eventually reverses. At the smallest configuration ($\Delta t = 0.5$ ms, MaxBS = 32), $L \approx 1.5$ –1.7 places both schedulers well below the hardware crossover, and MM’s GEMM setup overhead dominates (-38%). At the main experiment parameters ($\Delta t = 5$ ms, MaxBS = 128), L rises to 3.1–3.5 and the gap nearly closes (-2%). At $\Delta t = 20$ ms and MaxBS = 256, L reaches 5.8–7.3 and MM overtakes MV by 10%. The throughput gains come at a direct latency cost: MV average latency grows from 17.5 ms to 160.8 ms, while the sequential baseline maintains ~ 0.5 ms.

3.2.2 Clustered Workload. In contrast to the random workload, MM wins from small-to-medium batch sizes because clustered queries share many lists, keeping L above the crossover even in small batches. At $\Delta t = 0.5$ ms and MaxBS = 32, $L \approx 7.5$ –7.6 sits right at the hardware crossover and MM barely loses (-5%). As batch size grows, L rises well above the crossover and MM’s advantage increases steadily to $+24\%$.

Figure 2 summarises the sweep results across both workloads. The two trajectories—random rising from -38% and clustered starting near zero—illustrate how the query distribution over IVF clusters determines the operating regime of each scan mode.

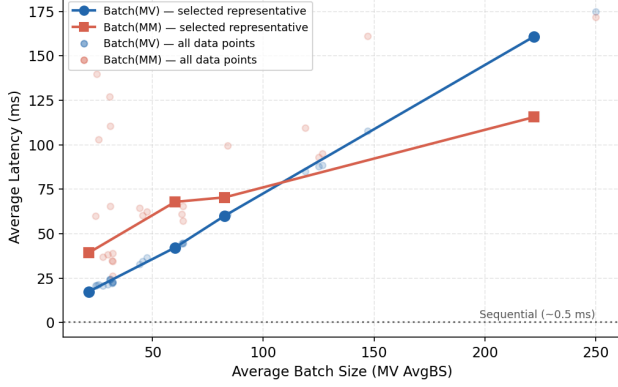
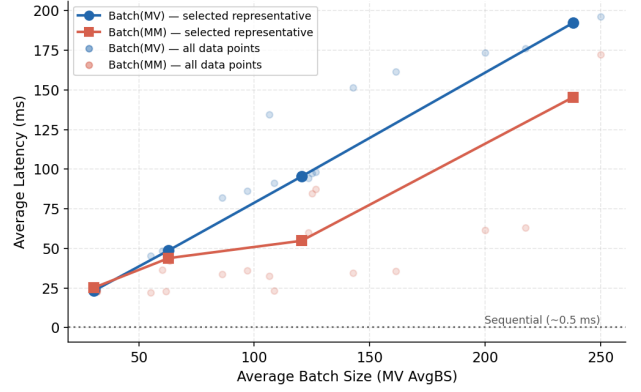
Figures 3 and 4 show the latency cost of batching. On the random workload, both schedulers incur substantially higher average latency than Sequential (~ 0.5 ms); at small batches MM latency exceeds MV’s due to GEMM setup overhead, but at large batches MM’s faster execution drains the queue sooner. On the clustered

Table 4: Selected (Δt , MaxBS) configurations – random workload.

Δt (ms)	MaxBS	MV QPS	MM QPS	Mean L (MV/MM)	MV Lat. (ms)	ΔQPS (MM–MV) (%)
0.5	32	2,098	1,300	1.5/1.7	17.5	–38%
5	64	2,230	1,446	2.5/2.5	42.3	–35%
5	128	2,277	2,227	3.1/3.5	60.1	–2%
20	256	2,322	2,562	7.3/5.8	160.8	+10%

Table 5: Selected (Δt , MaxBS) configurations – clustered workload.

Δt (ms)	MaxBS	MV QPS	MM QPS	Mean L (MV/MM)	MV Lat. (ms)	ΔQPS (MM–MV) (%)
0.5	32	1,987	1,884	7.5/7.6	23.2	–5%
5	64	1,934	2,156	11.1/11.0	48.7	+11%
5	128	1,952	2,324	14.6/11.9	95.4	+19%
20	256	1,948	2,424	17.8/17.3	192.3	+24%

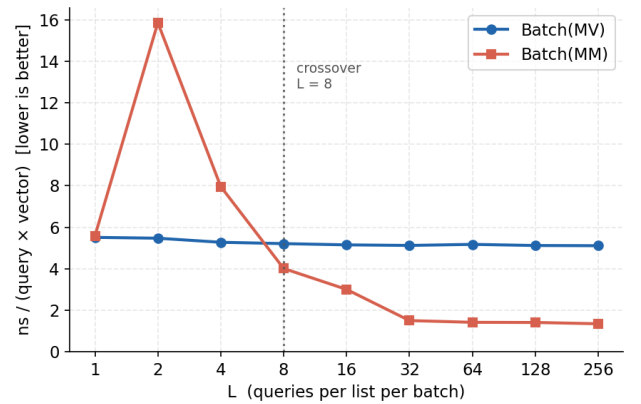
**Figure 3: Average latency vs batch size on the random workload.****Figure 4: Average latency vs batch size on the clustered workload.**

workload, MM latency runs consistently below MV because the high list-sharing rate gives MM a sustained throughput advantage.

3.3 Microbenchmark

To isolate the effect of L from scheduler dynamics, the MV and MM kernels are benchmarked directly on a synthetic inverted list ($n = 4,000$ vectors, $d = 128$) for $L \in \{1, \dots, 256\}$, with 50 repetitions per point. Figure 5 reports kernel cost in ns/(query $\times$ vector), so lower values indicate higher throughput.

MV throughput is approximately constant across all values of L (5.1–5.5 ns/q-v), as each query is processed independently. MM throughput improves sharply with L , crossing the MV baseline at $L = 8$ and reaching a 3.77 $\times$ speedup at $L = 256$. The crossover corresponds to the point where GEMM arithmetic intensity ($L/2$ FLOP/byte) exceeds the GEMV baseline (0.5 FLOP/byte). In the main experiment, mean $L \approx 3.6$ –4.2 on random (below the crossover) and mean $L \approx 12.4$ –14.6 on clustered (above it), fully accounting for the reversed MV–MM ordering observed in Table 2.

**Figure 5: Scan kernel throughput as a function of L .**

4 Discussion

The experiments show that batching is useful for IVF query execution: under the right workload and scan mode, batched execution can generate higher QPS than sequential processing.

Batching can improve throughput by reusing list work. The results indicate that time-window batching can improve throughput without modifying the index or changing Recall@10. Its benefit comes from reducing repeated list scans: unique list loads per query fall from 8 under Sequential to 2.23 on the random workload and 0.55 on the clustered workload. This gain is not free, because time-window batching introduces queueing and increases per-query latency; the saved scan work therefore must outweigh the latency cost.

The MV-MM choice is largely governed by L . Neither scan mode is uniformly best. Batch(MV) is stronger when each inverted list is shared by only a few queries, because it avoids the setup overhead of GEMM; Batch(MM) becomes stronger when enough queries share the same list. The microbenchmark provides evidence for this mechanism: MM overtakes MV around $L = 8$ on the experimental hardware, near the point where GEMM arithmetic intensity exceeds the GEMV baseline. Thus, in this setting, L —the number of queries within a batch that probe the same inverted list—provides the most direct explanation for scan-mode behavior.

L is shaped by batch size and query distribution. Larger batches place more queries in the same scheduling window, which makes it more likely that multiple queries probe a common inverted list. The query distribution over IVF clusters strengthens this effect by determining whether co-batched queries actually touch the same lists. This mechanism supports the batch-size sweep: increasing batch size and using more concentrated query distributions can both raise L and move the system toward an MM-favorable regime.

Practical Guideline. These results suggest that a scheduler should choose the scan mode from the L induced by its batch size and query distribution. A larger batch or a more concentrated query stream can increase per-list sharing, but the scheduler should rely on the observed L rather than either factor alone. Low per-list sharing favors Batch(MV), while high sharing favors Batch(MM). The crossover point between the two should be measured for the target hardware and workload.

Limitations and Future Work. The engine is deliberately single-threaded to isolate scheduling effects; whether the observed gains survive in a multi-core deployment remains an open question. An adaptive strategy that switches between MV and MM per batch based on observed L would eliminate the regression on random workloads while preserving MM’s advantage on clustered ones. All results are reported on SIFT1M; validation on datasets with different dimensionalities or cluster geometries would strengthen generalizability.

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